

MULTIANSWER QUESTIONS

49. As forces are in equilibrium we have

$$\vec{F}_1 + \vec{F}_2 + \vec{F}_3 + \vec{F}_4 = \vec{0}$$

equating the coefficients of $\hat{i}, \hat{j}, \hat{k}$ and solve the equations

50. Let point A be taken as origin. Then the position vectors of B, C and D are
- $\vec{a}, \vec{a} + \vec{b}$
- and
- $\vec{b} - \vec{a}$
- respectively

$$\text{P.V. of } M = \frac{\vec{a} + \vec{b} + \vec{a}}{2} = \vec{a} + \frac{\vec{b}}{2}$$

$$\Rightarrow DN : NM = 4 : 1$$

$$\text{P.V. of } N = \frac{4\left(\vec{a} + \frac{\vec{b}}{2}\right) + \vec{b} - \vec{a}}{5} = \frac{3}{5}(\vec{a} + \vec{b})$$

$$\vec{AN} = \frac{3}{5} \vec{AC}$$

51. Given that

$$3(4\vec{q} + 3\vec{r}) = 2\vec{q} - 3\vec{r}$$

$$\Rightarrow 10\vec{q} = -12\vec{r}$$

$$\Rightarrow \vec{q} = -\frac{6}{5}\vec{r} \dots (i)$$

$$(i) \vec{p} = 4\vec{q} + 3\left(-\frac{5}{6}\vec{q}\right) \quad (\text{by using } (i))$$

$$\Rightarrow \vec{p} = \frac{3}{2}\vec{q}$$

$$\Rightarrow |\vec{p}| > |\vec{q}| \text{ and directions of } \vec{p} \text{ and } \vec{q} \text{ are same}$$

$$(ii) \vec{p} = 4\left(-\frac{6}{5}\vec{r}\right) + 3\vec{r}$$

$$\vec{p} = -\frac{9}{5}\vec{r}$$

52. We have,

$$\alpha(\vec{i} - 3\vec{k}) + \beta(2\vec{i} + \vec{j}) + \gamma(\vec{i} - \vec{j} + \vec{k})$$

$$= -2\vec{i} - 5\vec{j} + 6\vec{k}$$

On comparing, we get

$$\alpha + 2\beta + \gamma = -2 \dots (i)$$

$$\beta - \gamma = -5 \dots (ii)$$

$$-3\alpha + \gamma = 6 \dots (iii)$$

on solving the equations (i), (ii) & (iii), we have

$$\alpha = -1, \beta = -2, \gamma = 3$$

$$53. \therefore \frac{\hat{a}}{|\hat{a}|} \cdot (\hat{a} + \hat{b}) = \frac{\hat{b}}{|\hat{b}|} \cdot (\hat{a} + \hat{b})$$

$$\text{or } \left(\frac{\vec{a}}{|\vec{a}|} - \frac{\vec{b}}{|\vec{b}|} \right) \cdot (\vec{a} + \vec{b}) = 0$$

which is possible when $\vec{a} = \vec{b}$

$$|\vec{a}| = |\vec{b}|$$

If angle between $\vec{a}$ and $\vec{b}$ is θ , then

$$\vec{a} \cdot \vec{a} = |\vec{a}| |\vec{a}| \cos \theta$$

$$\cos \theta = 1, \theta = 0^\circ$$

$$54. \cos^2 \theta + \sin^2 \theta + \sin^2 \theta = 1$$

$$\Rightarrow 3 \sin^2 \theta = 1 \Rightarrow \sin^2 \theta = \frac{1}{3}$$

$$\Rightarrow \sin \theta = \frac{1}{\sqrt{3}}$$

$$\tan \theta = \sqrt{2}$$

$$2) \vec{a} \cdot \vec{b} + \vec{c} = 0, \vec{b}(\vec{c} + \vec{a}) = 0, \vec{c}(\vec{a} + \vec{b}) = 0$$

$$\Rightarrow \vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a} = 0$$

$$|\vec{a} - \vec{b}|^2 = |\vec{a}|^2 + |\vec{b}|^2 - 2\vec{a} \cdot \vec{b}$$

$$= 9 + 16 - 2|\vec{a}| |\vec{b}| \sin \theta$$

$$|\vec{a} - \vec{b}|^2 = 25 - 2(3)(4) \sin \theta$$

$$= 25 - 24 \sin \theta$$

$$|\vec{a} - \vec{b}| = \sqrt{25 - 24 \sin \theta}$$

$$|\vec{a} - \vec{b}| = \sqrt{25 - 24} = 1$$

$$|\vec{a} - \vec{b}| = \sqrt{25 - 24} = 1$$

$$\text{range of } |\vec{a} - \vec{b}| \text{ is } [1, 7] = \sqrt{25 - 24} = 1$$

$$3) \text{ similarly we have calculate range of } |\vec{b} - \vec{c}|$$

VECTOR ALGEBRA

JEE ADVANCED - VOL - I

55. Let $\vec{c} = c_1\hat{i} + c_2\hat{j} + c_3\hat{k}$

$$\therefore c_1^2 + c_2^2 + c_3^2 = |\hat{i} + \hat{j}|^2 = 2$$

Also,

$$\frac{(c_1\hat{i} + c_2\hat{j} + c_3\hat{k}) \cdot (\hat{i} + \hat{j})}{2} = \frac{(c_1\hat{i} + c_2\hat{j} + c_3\hat{k})}{2} = \frac{1}{2}$$

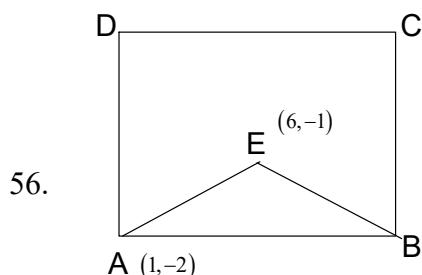
$$\Rightarrow c_1 + c_2 = c_2 + c_3 = 1 \Rightarrow c_1 = c_3 \text{ and}$$

$$c_2 = 1 - c_3$$

$$\text{Hence from (1), } 3c_3^2 - 2c_3 - 1 = 0 \Rightarrow c_3 = -\frac{1}{3}$$

$$\text{or } 1 \text{ when } c_3 = -\frac{1}{3}$$

$$\vec{c} = -\frac{1}{3}(\hat{i} - 4\hat{j} + \hat{k}) \text{ when } c_3 = 1, \vec{c} = \hat{i} + \hat{k}$$



E is the mid-point of A and C

Position vector of C is $11\hat{i}$

$$|\underline{AEB}| = |\underline{DEA}| = \frac{\pi}{2}$$

$$AB^2 = AD^2 = 2AE^2$$

If B or D is (x, y)

$$\overrightarrow{BE} \cdot \overrightarrow{AE} = 0$$

$$\text{and } \overrightarrow{DE} \cdot \overrightarrow{AE} = 0$$

$$\Rightarrow 5x + y = 29$$

$$\text{and } (x-1)^2 + (y+2)^2 = 52$$

$$\Rightarrow x^2 - 12x + 35 = 0$$

$$\Rightarrow x = 5, y = 4$$

$$x = 7, y = -6$$

$$\overrightarrow{OB} = 7\hat{i} - 6\hat{j} \quad \overrightarrow{OD} = 5\hat{i} + 4\hat{j} \text{ and area} = 52 \text{ sq. units}$$

57. According to the given conditions $\mathbf{a} \cdot \mathbf{b} > 0$ and

$\mathbf{b} \cdot \mathbf{c} < 0$, where $\mathbf{c} = (0, 1, 0)$. Thus

$2x^2 - 3x + 1 > 0$ for $x < 0$. But

$2x^2 - 3x + 1$ is greater than zero for all $x < 0$

58. Let $\mathbf{b} = x\mathbf{i} + y\mathbf{j}$. Since $\mathbf{a}$ is perpendicular to $\mathbf{b}$ so

$$4x + 3y = 0. \text{ Thus } \mathbf{b} = x\left(\mathbf{i} - \frac{4}{3}\mathbf{j}\right).$$

Let $\mathbf{c} = u\mathbf{i} + v\mathbf{j}$ be the required vector. According to the given condition

$$1 = \frac{\mathbf{c} \cdot \mathbf{a}}{|\mathbf{a}|} \Rightarrow 4u + 3v = 5.$$

$$\text{Also } 2 = \frac{\mathbf{c} \cdot \mathbf{b}}{|\mathbf{b}|} \Rightarrow \frac{ux - (4/3)vx}{\sqrt{x^2(1 + (16/9))}} = 2$$

$$\Rightarrow 3u - 4v = \pm 10$$

Solving these equations we have

$$u = 2 \text{ and } v = -1 \text{ or } u = -2/5, v = 11/5$$

59. The diagonals are given by

$$\mathbf{AB} - \mathbf{BC} = 4\mathbf{i} - 2\mathbf{j} + 4\mathbf{k}, \mathbf{AB} + \mathbf{BC} = 2\mathbf{i} - 2\mathbf{j}$$

These vectors have magnitudes 6 and $2\sqrt{2}$, respectively, and their dot product is 12.

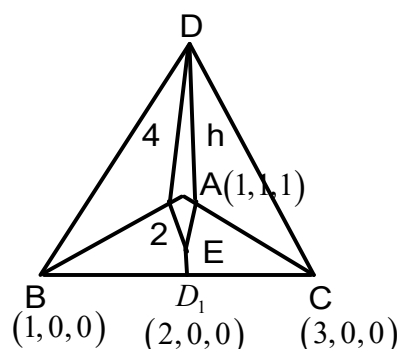
Therefore the angle between them is

$$\cos^{-1} \frac{12}{(6)(2\sqrt{2})} = \cos^{-1} \frac{1}{\sqrt{2}} = \frac{\pi}{4} \text{ or } \frac{3\pi}{4}$$

60. $\triangle ADE$ is right angled at E

$$v = (h \times \text{Area } \triangle ABC) \frac{1}{3}$$

$$\frac{2\sqrt{2}}{3} h \times \frac{1}{2} \times |\overrightarrow{AB} \times \overrightarrow{AC}|$$



$$\overrightarrow{AB} = -\vec{j} - \vec{k}, \quad \overrightarrow{AC} = 2\vec{i} - \vec{j} - \vec{k}$$

$$\overrightarrow{AB} \times \overrightarrow{AC} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 0 & -1 & -1 \\ 2 & -1 & -1 \end{vmatrix}$$

$$= \vec{i}(0) - \vec{j}(2) + \vec{k}(2)$$

$$= 2\vec{k} - 2\vec{j}$$

$$|\overrightarrow{AB} \times \overrightarrow{AC}| = \sqrt{4+4} = 2\sqrt{2}$$

$$\therefore \frac{2\sqrt{2}}{3} = \left(h \times \frac{1}{2} \times 2\sqrt{2} \right) \frac{1}{3}$$

$$\Rightarrow h = 2$$

$$AE = \sqrt{16-4} = 2\sqrt{3}$$

$$AD_1 = \sqrt{1+1+1} = \sqrt{3}$$

E divide AD_1 in the ratio

$$2\sqrt{3} : \sqrt{3} \Rightarrow 2:1 \text{ externally}$$

then P.V if B is $3\vec{i} - \vec{j} - \vec{k}$

or 2:3 externally

P.V.E. $-\vec{i} + 3\vec{j} + 3\vec{k}$.

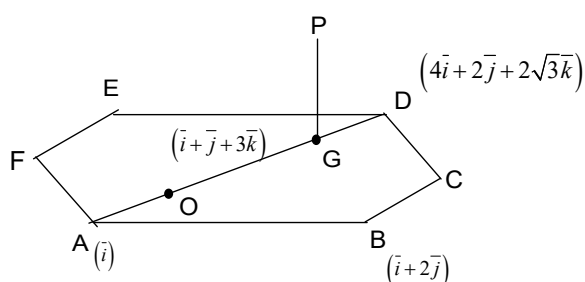
61. Since vectors $(x, x+1, x+2)$, $(x+3, x+4, x+5)$ and $(x+6, x+7, x+8)$ are coplanar

$$\therefore \begin{vmatrix} x & x+1 & x+2 \\ x+3 & x+4 & x+5 \\ x+6 & x+7 & x+8 \end{vmatrix} = 0$$

Applying $C_2 \rightarrow C_2 - C_1$ and $C_3 \rightarrow C_3 - C_1$

$$\therefore \begin{vmatrix} x & 1 & 2 \\ x+3 & 1 & 2 \\ x+6 & 1 & 2 \end{vmatrix} = 0$$

62. Let the centre of the base be (O)



$$\overrightarrow{AB} = 2\vec{i} \Rightarrow |\overrightarrow{AB}| = 2$$

$$\text{Area of the } \Delta OAB = \frac{1}{4} \cdot 4 \cdot \sqrt{3} = \sqrt{3}$$

$$\Rightarrow \text{Base area} = 6\sqrt{3} \text{ sq. units}$$

Let height of the pyramid be h.

$$\Rightarrow \frac{1}{3} \cdot 6\sqrt{3}h = 6\sqrt{3} \Rightarrow h = 3 \text{ units}$$

$$\overrightarrow{AP} = 3\vec{i} + 2\vec{j} + 2\sqrt{3}\vec{k}$$

$$|\overrightarrow{AP}| = 5 \text{ units}, \quad |\overrightarrow{AG}| = 4 \text{ units}$$

Now, $\overrightarrow{AG}$ and $\overrightarrow{AO}$ are collinear

$$\overrightarrow{AG} = \lambda \overrightarrow{AO} = |\overrightarrow{AG}| = |\lambda| |\overrightarrow{AO}|$$

$$\Rightarrow 2|\lambda| = 4 \Rightarrow |\lambda| = 2$$

$$\overrightarrow{AG} = \pm 2(\vec{i} + \vec{j} + \sqrt{3}\vec{k})$$

$$\vec{G} = \pm 2(\vec{i} + \vec{j} + \sqrt{3}\vec{k}) + \vec{i}$$

$$= 2\vec{i} + 2\vec{j} + 2\sqrt{3}\vec{k} + \vec{i},$$

$$= -2\vec{i} - 2\vec{j} - 2\sqrt{3}\vec{k} + \vec{i}$$

$$= 3\vec{i} + 2\vec{j} + 2\sqrt{3}\vec{k}, -\vec{i} - 2\vec{j} - 2\sqrt{3}\vec{k}$$

63. $\vec{a} \cdot \vec{r} = \vec{a} \cdot \vec{b} \dots \dots \dots (1)$

$$\vec{a} \times \vec{r} + \vec{a} \times (\vec{r} \times \vec{a}) = \vec{a} \times \vec{b}.$$

$$\vec{r} - \vec{b} + (\vec{a} \cdot \vec{a})\vec{r} - (\vec{a} - \vec{b})\vec{a} = \vec{a} \times \vec{b}$$

$$(1 + \vec{a} \cdot \vec{a})\vec{r} = (\vec{a} \cdot \vec{b})\vec{a} + \vec{b} + \vec{a} \times \vec{b}$$

$$\vec{r} = \frac{(\vec{a} \cdot \vec{b})\vec{a} + \vec{b} + \vec{a} \times \vec{b}}{1 + \vec{a} \cdot \vec{a}}$$

and also find $|\vec{r}|$

64. Since $\vec{a}, \vec{b}$ and $\vec{c}$ are unit vectors inclined at an angle θ .

$$|\vec{a}| = |\vec{b}| = 1 \text{ and } \cos \theta = \vec{a} \cdot \vec{c} = \vec{b} \cdot \vec{c}$$

$$\text{Now, } \vec{c} = \alpha \vec{a} + \beta \vec{b} + \gamma (\vec{a} \times \vec{b}) \quad (1)$$

$$\Rightarrow \vec{a} \cdot \vec{c} = \alpha (\vec{a} \cdot \vec{a}) + \beta (\vec{a} \cdot \vec{b}) + \gamma \{ \vec{a} \cdot (\vec{a} \times \vec{b}) \}$$

$$\Rightarrow \cos \theta = \alpha |\vec{a}|^2$$

$$[\because \vec{a} \cdot \vec{b} = 0, \vec{a} \cdot (\vec{a} \times \vec{b}) = 0]$$

$$\Rightarrow \cos \theta = \alpha$$

Similarly, by taking dot product on both sides of

(1) by $\vec{b}$, we get $\beta = \cos \theta$

$$\therefore \alpha = \beta$$

$$\text{Again, } \vec{c} = \alpha \vec{a} + \beta \vec{b} + \gamma (\vec{a} \times \vec{b})$$

$$\Rightarrow |\vec{c}|^2 = |\alpha \vec{a} + \beta \vec{b} + \gamma (\vec{a} \times \vec{b})|^2$$

$$= \alpha^2 |\vec{a}|^2 + \beta^2 |\vec{b}|^2 + \gamma^2 |\vec{a} \times \vec{b}|^2 + 2\alpha\beta (\vec{a} \cdot \vec{b})$$

$$+ 2\alpha\gamma \{ \vec{a} \times \vec{b} \} + 2\beta\gamma \{ \vec{b} \cdot (\vec{a} \times \vec{b}) \}$$

$$\Rightarrow 1 = \alpha^2 + \beta^2 + \gamma^2 |\vec{a} \times \vec{b}|^2$$

$$\Rightarrow 1 = 2\alpha^2 + \gamma^2 \left\{ |\vec{a}|^2 |\vec{b}|^2 \sin^2 \frac{\pi}{2} \right\}$$

$$\Rightarrow 1 = 2\alpha^2 + \gamma^2 \Rightarrow \alpha^2 = \frac{1 - \gamma^2}{2}$$

But $\alpha = \beta = \cos \theta$.

$$1 = 2\alpha^2 + \gamma^2 \Rightarrow \gamma^2 = 1 - 2\cos^2 \theta = -\cos 2\theta$$

$$\therefore \beta^2 = \frac{1 - \gamma^2}{2} = \frac{1 + \cos 2\theta}{2}$$

65. We have,

$$\text{AM} = \text{projection of } \vec{b} \text{ on } \vec{a} = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}|}$$

$$\therefore \overrightarrow{AM} = \left(\frac{\vec{a} \cdot \vec{b}}{|\vec{a}|^2} \right) \vec{a}$$

Now, in $\triangle ADM$

$$\overrightarrow{AD} = \overrightarrow{AM} + \overrightarrow{MD} \Rightarrow \overrightarrow{DM} = \overrightarrow{AM} - \overrightarrow{AD}$$

$$\Rightarrow \overrightarrow{DM} = \frac{(\vec{a} \cdot \vec{b}) \vec{a}}{|\vec{a}|^2} - \vec{b}$$

$$\text{Also, } \overrightarrow{DM} = \frac{1}{|\vec{a}|^2} [(\vec{a} \cdot \vec{b}) \vec{a} - |\vec{a}|^2 \vec{b}]$$

$$\Rightarrow \overrightarrow{MD} = \frac{1}{|\vec{a}|^2} [|\vec{a}|^2 \vec{b} - (\vec{a} \cdot \vec{b}) \vec{a}]$$

Now,

$$\frac{\vec{a} \times (\vec{a} \times \vec{b})}{|\vec{a}|^2} = \frac{1}{|\vec{a}|^2} [(\vec{a} \cdot \vec{b}) \vec{a} - (\vec{a} \cdot \vec{a}) \vec{b}] = \overrightarrow{DM}$$

$$66. \quad \vec{a} \times (\vec{r} \times \vec{a}) = \vec{a} \times \vec{b}$$

$$3\vec{r} - (\vec{a} \cdot \vec{r}) \vec{a} = \vec{a} \times \vec{b}$$

$$\text{Also } |\vec{r} \times \vec{a}| = |\vec{b}|$$

$$\Rightarrow \sin^2 \theta = 2/3$$

$$\Rightarrow (1 - \cos^2 \theta) = \frac{2}{3}$$

$$\Rightarrow \frac{1}{3} = \cos^2 \theta \quad \Rightarrow \vec{a} \cdot \vec{r} = \pm 1$$

$$\Rightarrow 3\vec{r} \pm \vec{a} = \vec{a} \times \vec{b} \quad \Rightarrow \vec{r} = \frac{1}{3} (\vec{a} \times \vec{b} \pm \vec{a})$$

$$67. \quad \text{Given } \vec{c} = \lambda_1 \vec{a} + \lambda_2 \vec{b} + \lambda_3 (\vec{a} \times \vec{b})$$

$$\text{and } \vec{a} \cdot \vec{b} = 0, |\vec{a}| = 1, |\vec{b}| = 1$$

$$\text{From (1), } \vec{a} \cdot \vec{c} = \lambda_1, \vec{c} \cdot \vec{b} = \lambda_2$$

$$\text{and } \vec{c} \cdot (\vec{a} \times \vec{b}) = |\vec{a} \times \vec{b}|^2 \lambda_3 \\ = (1 \cdot 1 \sin 90^\circ)^2 \lambda_3 = \lambda_3$$

$$\text{Hence } \lambda_1 + \lambda_2 + \lambda_3 = (\vec{a} + \vec{b} + \vec{a} \times \vec{b}) \cdot \vec{c}$$

$$68. \quad \text{Let } \vec{a} = -\frac{3}{2}(\hat{i} - \sqrt{3}\hat{j}), \vec{b} = \frac{3}{2}(\hat{i} - \sqrt{3}\hat{j}),$$

$$\vec{c} = -\frac{3}{2}(\hat{i} + \sqrt{3}\hat{j}), \vec{d} = \frac{3}{2}(\hat{i} + \sqrt{3}\hat{j})$$

$$\overrightarrow{AB} = 3\hat{i} = \vec{a} \quad (\text{say})$$

$$\text{Clearly, } |\vec{a}| = |\vec{b}| = |\vec{c}| = |\vec{d}| = 3$$

If $\vec{a}$ makes angle α, β and γ and δ with $\vec{a}, \vec{b}, \vec{c}$ and $\vec{d}$, respectively, then

$$\cos \alpha = \frac{\vec{a} \cdot \vec{a}}{|\vec{a}| |\vec{a}|} = \frac{-9}{3 \cdot 3} = -\frac{1}{2}$$

$$\cos \beta = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}| |\vec{b}|} = \frac{\frac{9}{2}}{3 \times 3} = \frac{1}{2} \Rightarrow \beta = 60^\circ$$

$$\cos \gamma = \frac{\vec{a} \cdot \vec{c}}{|\vec{a}| |\vec{c}|} = \frac{-9}{3 \times 3} = -\frac{1}{2} \Rightarrow \gamma = 120^\circ$$

$$\cos \delta = \frac{\vec{a} \cdot \vec{d}}{|\vec{a}| |\vec{d}|} = \frac{\frac{9}{2}}{3 \times 3} = \frac{1}{2} \Rightarrow \delta = 60^\circ$$

$$69. \quad \vec{d} \cdot \vec{a} = [\vec{a} \ \vec{b} \ \vec{c}] \cos y = -\vec{d} \cdot (\vec{b} + \vec{c})$$

$$\Rightarrow \cos y = \frac{\vec{d} \cdot (\vec{b} + \vec{c})}{[\vec{a} \ \vec{b} \ \vec{c}]}$$

Similarly,

$$\sin x = -\frac{\vec{d} \cdot (\vec{a} + \vec{b})}{[\vec{a} \ \vec{b} \ \vec{c}]} \text{ and } \frac{\vec{d} \cdot (\vec{a} + \vec{c})}{[\vec{a} \ \vec{b} \ \vec{c}]} = -2$$

$$\therefore \sin x + \cos y + 2 = 0$$

$$\Rightarrow \sin x + \cos y = -2$$

$$\Rightarrow \sin x = -1, \cos y = -1$$

Since we want the minimum value of $x^2 + y^2$ is

$$\frac{5\pi^2}{4}$$

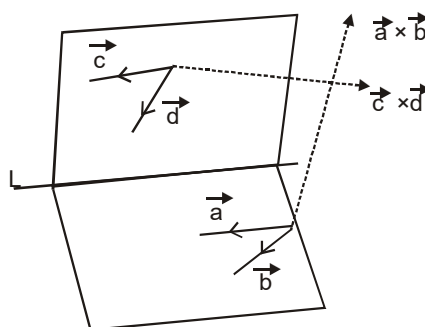
$$70. \quad \text{Here } (l\vec{a} + m\vec{b}) \times \vec{b} = \vec{c} \times \vec{b} \Rightarrow l\vec{a} \times \vec{b} = \vec{c} \times \vec{b}$$

$$\Rightarrow l(\vec{a} \times \vec{b})^2 = (\vec{c} \times \vec{b}) \cdot (\vec{a} \times \vec{b})$$

$$\Rightarrow l = \frac{(\vec{c} \times \vec{b}) \cdot (\vec{a} \times \vec{b})}{(\vec{a} \times \vec{b})^2}$$

$$\text{Similarly, } m = \frac{(\vec{c} \times \vec{a}) \cdot (\vec{b} \times \vec{a})}{(\vec{b} \times \vec{a})^2}$$

71.



$(\vec{a} \times \vec{b}) \times (\vec{c} \times \vec{d})$ is a vector perpendicular to $\vec{a} \times \vec{b}$ and $\vec{c} \times \vec{d}$.

If l is the line of intersection of the two planes of $\vec{a}, \vec{b}$ and $\vec{c}, \vec{d}$ then $l \perp \vec{a} \times \vec{b}$

and $l \perp \vec{c} \times \vec{d}$. So, $(\vec{a} \times \vec{b}) \times (\vec{c} \times \vec{d})$ is also l which is equally inclined to $\vec{a} \times \vec{b}$ and $\vec{c} \times \vec{d}$.

ASSERTION-REASONING QUESTIONS

72. Conceptual

73. Conceptual

74. conceptual

$$75. \quad 3\vec{a} - 2\vec{b} + 5\vec{c} - 6\vec{d}$$

$$= (2\vec{a} - 2\vec{b}) + (-5\vec{a} + 5\vec{c}) + (6\vec{a} - 6\vec{d})$$

$$76. \quad \text{We have adjacent sides of triangle } |\vec{a}| = 3, |\vec{b}| = 4$$

$$\text{The length of the diagonal is } |\vec{a} + \vec{b}| = 5$$

Since it satisfies the Pythagoras theorem, $\vec{a} \perp \vec{b}$

Hence the parallelogram is rectangle

$$77. \quad [\vec{b} \times \vec{c}, \vec{c} \times \vec{a}, \vec{a} \times \vec{b}] = [\vec{a} \ \vec{b} \ \vec{c}]^2 \neq 0$$

because by Statement I $\vec{a}, \vec{b}, \vec{c}$ are non-coplanar

78. Let

$$\vec{a} = a_1\vec{i} + a_2\vec{j} + a_3\vec{k}$$

$$\vec{b} = b_1\vec{i} + b_2\vec{j} + b_3\vec{k}$$

$$\vec{c} = c_1\vec{i} + c_2\vec{j} + c_3\vec{k}$$

Then

$$[\vec{a}\vec{b}\vec{c}](\vec{a} \times \vec{b}) = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

If A and B are two matrices of 3×3 order, then we know that $\det(AB) = (\det A)(\det B)$. Therefore

$$[\vec{a}\vec{b}\vec{c}](\vec{a} \times \vec{b}) = \begin{vmatrix} \vec{a} \cdot \vec{a} & \vec{a} \cdot \vec{b} & \vec{a} \cdot \vec{c} \\ \vec{b} \cdot \vec{a} & \vec{b} \cdot \vec{b} & \vec{b} \cdot \vec{c} \\ \vec{c} \cdot \vec{a} & \vec{c} \cdot \vec{b} & \vec{c} \cdot \vec{c} \end{vmatrix}$$

$\vec{a}, \vec{b}, \vec{c}$ are coplanar

$$\Rightarrow [\vec{a}\vec{b}\vec{c}] = 0. \text{ So } [\vec{a}\vec{b}\vec{c}](\vec{a} \times \vec{b}) = \vec{0}$$

79. According to Theorem 6.44, Statement II is correct.

In the given lines

$$\vec{a} = (3, 5, 7), \vec{b} = (1, -2, 1), \vec{c} = (-1, -1, -1)$$

$$\text{and } \vec{d} = (7, -6, 1). \text{ Now } \vec{a} - \vec{c} = (4, 6, 8)$$

$$\text{and } \vec{b} \times \vec{d} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & -2 & 1 \\ 7 & -6 & 1 \end{vmatrix} = 4\vec{i} + 6\vec{j} + 8\vec{k}$$

Therefore the shortest distance is

$$\frac{|(\vec{a} - \vec{c}) \cdot (4\vec{i} + 6\vec{j} + 8\vec{k})|}{2\sqrt{4 + 9 + 16}} = \frac{|16 + 36 + 64|}{2\sqrt{29}} = \frac{58}{\sqrt{29}} = 2\sqrt{29}$$

Hence Statement I is false

80. Statement II is clearly true. It is known that the angles

$$(\vec{OB}, \vec{OC}) = 2A, (\vec{OC}, \vec{OA}) = 2B$$

$$\text{and } (\vec{OA}, \vec{OB}) = 2C$$

$$0 \leq |\vec{OA} + \vec{OB} + \vec{OC}|^2$$

$$= |\vec{OA}|^2 + |\vec{OB}|^2 + |\vec{OC}|^2 + 2 \sum (\vec{OB} \cdot \vec{OC})$$

$$= 3R^2 + 2R^2(\cos 2A + \cos 2B + \cos 2C)$$

where $OA = OB = OC = R$. Therefore

$$0 \leq 3 + 2(\cos 2A + \cos 2B + \cos 2C)$$

Hence

$$-\frac{3}{2} \leq \cos 2A + \cos 2B + \cos 2C$$

81. We have

$$\vec{PQ} \times (\vec{RS} + \vec{ST}) = \vec{PQ} \times \vec{RT} \neq \vec{0}$$

because $\vec{PQ}$ and $\vec{RT}$ are not parallel.

Statement I is true. Now

$$\vec{PQ} \times \vec{RS} \neq \vec{0}$$

because they are not parallel vectors. Also

$$\vec{PQ} \times \vec{ST} = \vec{0} \text{ because } \vec{PQ} \text{ and } \vec{ST} \text{ are parallel.}$$

Hence Statement II is false.

Answer: (C)

82. We have

$$\vec{n}_1 \times \vec{n}_2 = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 3 & -6 & -2 \\ 2 & 1 & -2 \end{vmatrix} = 14\vec{i} + 2\vec{j} + 15\vec{k}$$

83. $(6 - \lambda)\vec{a} + (2\lambda - 1)\vec{c} = \vec{a}(1 + \mu) + \vec{c}(3\mu - 1)$

$$\Rightarrow 6 - \lambda = 1 + \mu$$

$$\lambda + \mu = 5 \quad \dots\dots\dots(1)$$

$$\Rightarrow 2\lambda - 1 = 3\mu - 1$$

$$2\lambda = 3\mu \quad \dots\dots\dots(2)$$

Solving (1) & (2) two unique value of λ and μ can be found out for which both the $\vec{r}$ will be same

84. In an isosceles triangle ABC in which $AB = AC$, the median and bisector from A must be same line

$\Rightarrow$ reason R is true.

$$\text{Now } \vec{AD} = \frac{\vec{u} + \vec{v}}{2} \text{ and}$$

$$|\vec{AD}|^2 = \frac{1}{4} [|\vec{u}|^2 + |\vec{v}|^2 + 2\vec{u} \cdot \vec{v}]$$

$$= \frac{1}{4} [1 + 1 + 2 \cos \alpha] = \frac{1}{2} \cdot 2 \cos^2 \frac{\alpha}{2}$$

$$\Rightarrow |\vec{AD}| = \cos \frac{\alpha}{2}$$

$\Rightarrow$ unit vector along AD i.e., $\hat{x}$ is given by

$$\vec{x} = \frac{\overrightarrow{AD}}{|\overrightarrow{AD}|} = \frac{\vec{u} + \vec{v}}{2\cos\frac{\alpha}{2}}$$

85. Statement 1 is wrong since by using

$$[\vec{a} \vec{b} \vec{c}] = \vec{a} \cdot (\vec{b} \times \vec{c})$$

We get

$$[2\vec{a} + 3\vec{b}, 3\vec{b} + 5\vec{c}, 5\vec{c} + 7\vec{a}] = 135[\vec{a} \vec{b} \vec{c}]$$

Statement 1 is false, statement 2 is true

86. Both statement 1 and 2 are true and statement 2 is correct explanation of statement 1
statement 2 is indeed correct explanation of statement 1

PASSAGE QUESTIONS

87. Let $\overrightarrow{OA} = \vec{a}$, $\overrightarrow{OB} = \vec{b}$, $\overrightarrow{OC} = \vec{c}$ be conterminus edges of a tetrahedron (regular)

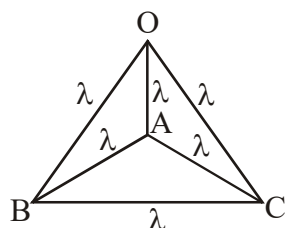
$$|\vec{a}| = |\vec{b}| = |\vec{c}| = \lambda \text{ the length of the edge.}$$

Normal to the face ABC is $\vec{n} = \overrightarrow{AB} \times \overrightarrow{AC}$

$$= \vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a}$$

Angle between $\vec{a}$ and $\vec{n}$ is θ

$$= \cos \theta = \frac{\vec{a} \cdot \vec{n}}{|\vec{a}| |\vec{n}|}$$



$$= \frac{(\vec{a} \cdot \vec{n})}{|\vec{a}| |\overrightarrow{AB} \times \overrightarrow{AC}|} = \frac{(\lambda^6)^{1/2}}{\lambda |\overrightarrow{AB}| |\overrightarrow{AC}|} = \sqrt{\frac{2}{3}}$$

required angle is $90 - \theta$

$$\Rightarrow \cos(90 - \theta) = \sin \theta = \sqrt{1 - \frac{2}{3}} = \frac{1}{\sqrt{3}}$$

$$\therefore 90 - \theta = \cos^{-1}\left(\frac{1}{\sqrt{3}}\right)$$

$$[\vec{a} \vec{b} \vec{c}]^2 = \begin{vmatrix} \lambda^2 & \frac{\lambda^2}{2} & \frac{\lambda^2}{2} \\ \frac{\lambda^2}{2} & \lambda^2 & \frac{\lambda^2}{2} \\ \frac{\lambda^2}{2} & \frac{\lambda^2}{2} & \lambda^2 \end{vmatrix}$$

$$= \lambda^6 \begin{vmatrix} 1 & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & 1 & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & 1 \end{vmatrix}$$

$$= \lambda^6 \left\{ \left(1 - \frac{1}{4}\right) - \frac{1}{2} \left(\frac{1}{2} - \frac{1}{4}\right) + \frac{1}{2} \left(\frac{1}{4} - \frac{1}{2}\right) \right\}$$

$$= \lambda^6 \left(\frac{3}{4} - \frac{1}{8} - \frac{1}{8} \right)$$

$$= \lambda^6 \left(\frac{3}{4} - \frac{1}{4} \right) = \frac{\lambda^6}{2}$$

$$[\vec{a} \vec{b} \vec{c}] = \frac{\lambda^3}{\sqrt{2}}$$

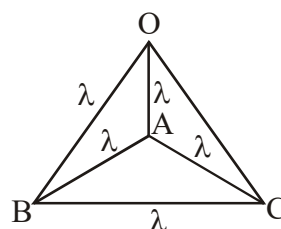
88. Consider 2 edges

$$\overrightarrow{OA} = \vec{a}$$

$$\text{and } \overrightarrow{BC} = \vec{c} - \vec{b}$$

Let θ be the angle between opposite edges

$$\cos \theta = \frac{\overrightarrow{OA} \cdot \overrightarrow{BC}}{|\overrightarrow{OA}| |\overrightarrow{BC}|}$$



$$= \frac{\vec{a} \cdot (\vec{c} - \vec{b})}{|\vec{a}| |\overrightarrow{BC}|}$$

$$= \frac{\vec{a} \cdot \vec{c} - \vec{a} \cdot \vec{b}}{\lambda^2} = 0$$

$$\theta = 90^\circ$$

VECTOR ALGEBRA

JEE ADVANCED - VOL - I

89. Equation of plane ABC in the above diagram is

$$\vec{r} \cdot \frac{\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a}}{|\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a}|} = \frac{[\vec{a} \ \vec{b} \ \vec{c}]}{|\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a}|}$$

⊥ distance from vertex O to the plane ABC

$$\frac{[\vec{a} \ \vec{b} \ \vec{c}]}{|\vec{AB} \times \vec{AC}|} = \frac{\lambda^3}{\sqrt{2} |\vec{AB}| |\vec{AB}| \sin 60^\circ}$$

$$= \frac{\lambda^3}{\frac{\sqrt{2}}{\lambda^2 \cdot \frac{\sqrt{3}}{2}}} = \sqrt{\frac{2}{3}} \lambda$$

here $\lambda = a$

⊥ distance from vertex to the opposite face =

$$\sqrt{\frac{2}{3}} a$$

90, 91, 92.

$$\vec{a} \times \vec{b} + \vec{b} \times \vec{c} = p\vec{a} + q\vec{b} + r\vec{c} \quad \dots (1)$$

taking dot product with $\vec{a}, \vec{b}, \vec{c}$ respectively in (1)

$$[\vec{a} \ \vec{b} \ \vec{c}] = p + q \cos \theta + r \cos \theta \quad \dots (2)$$

$$0 = p \cos \theta + q + r \cos \theta \quad \dots (3)$$

$$[\vec{a} \ \vec{b} \ \vec{c}] = p \cos \theta + q \cos \theta + r \quad \dots (4)$$

(2) + (3) + (4) gives :

$$p + q + r = \frac{2[\vec{a} \ \vec{b} \ \vec{c}]}{2 \cos \theta + 1} \quad \dots (5)$$

(2) - (5) $\times \cos \theta$ gives

$$p = \frac{[\vec{a} \ \vec{b} \ \vec{c}]}{(2 \cos \theta + 1)(1 - \cos \theta)} = r$$

$$q = \frac{-2[\vec{a} \ \vec{b} \ \vec{c}] \cos \theta}{(2 \cos \theta + 1)(1 - \cos \theta)}$$

$$\text{But } [\vec{a} \ \vec{b} \ \vec{c}] = \begin{vmatrix} 1 & \cos \theta & \cos \theta \\ \cos \theta & 1 & \cos \theta \\ \cos \theta & \cos \theta & 1 \end{vmatrix}$$

$$= (1 + 2 \sin \theta)(1 - \sin \theta)^2$$

$$p = r = \frac{1}{\sqrt{1 + 2 \cos \theta}}, \quad q = \frac{-2 \cos \theta}{\sqrt{1 + 2 \cos \theta}}$$

93, 94, 95.

$\vec{u} + \vec{v} + \vec{w} = \vec{a}$ taking dot product with $\vec{u}$ on both sides.

$$1 + \vec{u} \cdot \vec{v} + \vec{u} \cdot \vec{w} = \vec{a} \cdot \vec{u} = \frac{3}{2}$$

$$\vec{u} \cdot \vec{v} + \vec{u} \cdot \vec{w} = \frac{1}{2} \quad \dots (1)$$

similarly by taking dot product with v and w

$$\text{we get } \vec{u} \cdot \vec{v} + \vec{u} \cdot \vec{w} = \frac{3}{4} \quad \dots (2)$$

$$\vec{u} \cdot \vec{w} + \vec{v} \cdot \vec{w} = \frac{-1}{4} \quad \dots (3)$$

$$(1) + (2) + (3) \quad \vec{u} \cdot \vec{v} + \vec{v} \cdot \vec{w} + \vec{w} \cdot \vec{u} = \frac{1}{2} \quad \dots (4)$$

from the above (1), (2), (3), (4) we get

$$\vec{v} \cdot \vec{w} = 0, \quad \vec{u} \cdot \vec{w} = -\frac{1}{4}, \quad \vec{u} \cdot \vec{v} = \frac{3}{4}$$

Now by using $\vec{u} \times (\vec{v} \times \vec{w}) = \vec{b}$

$$(\vec{u} \times \vec{v}) \times \vec{w} = \vec{c}$$

$$\text{we get } \vec{v} = -4\vec{c}, \quad \vec{w} = \frac{4}{3}(\vec{c} - \vec{b})$$

$$\vec{u} = \vec{a} + \frac{4}{3}\vec{b} + \frac{8}{3}\vec{c}$$

96, 97, 98.

$\vec{u} \times \vec{v} = \vec{b}$ taking cross product by $\vec{a}$

$$\vec{a} \times (\vec{u} \times \vec{v}) = \vec{a} \times \vec{b}$$

$$(\vec{a} \cdot \vec{v})\vec{u} - (\vec{a} \cdot \vec{u})\vec{v} = \vec{a} \times \vec{b}$$

$$\vec{u} - \vec{v} = \vec{a} \times \vec{b} \rightarrow (1)$$

By takes cross product with $\vec{a}$ on both sides

$$\vec{v} \times \vec{w} = \vec{c}$$

$$\vec{a} \times (\vec{v} \times \vec{w}) = \vec{a} \times \vec{c}$$

$$(\vec{a} \cdot \vec{w})\vec{v} - (\vec{a} \cdot \vec{v})\vec{w} = \vec{a} \times \vec{c} \rightarrow (2)$$

Taking dot product in

$$\vec{u} + \vec{v} + \vec{w} = \vec{a} \text{ by } \vec{a} \text{ on both sides}$$

$$\vec{a} \cdot \vec{u} + \vec{a} \cdot \vec{v} + \vec{a} \cdot \vec{w} = \vec{a} \cdot \vec{a} = 3$$

$$\vec{a} \cdot \vec{w} = 1$$

$$\vec{v} - \vec{w} = \vec{a} \times \vec{c}$$

$$\text{Solving } \vec{u} + \vec{v} + \vec{w} = \vec{a}, (1) \text{ and } (2)$$

We get

$$\vec{u} = \frac{\vec{a} + \vec{a} \times \vec{c} + 2\vec{a} \times \vec{b}}{3}$$

$$\vec{v} = \frac{\vec{a} + \vec{a} \times \vec{c} - \vec{a} \times \vec{b}}{3}$$

$$\vec{w} = \frac{\vec{a} - 2\vec{a} \times \vec{c} - \vec{a} \times \vec{b}}{3}$$

MATRIX MATCHING QUESTIONS

99. (A) A vector l/r to the line is $\vec{b} = \vec{i} + 2\vec{j} + 3\vec{k}$

$$\therefore \text{unit vector along the line is } \frac{\vec{i} + 2\vec{j} + 3\vec{k}}{\sqrt{14}}$$

$\therefore$ The projection =

$$(\vec{i} + \vec{j} + \vec{k}) \cdot \frac{(\vec{i} + 2\vec{j} + 3\vec{k})}{\sqrt{14}} = \frac{6}{\sqrt{14}}$$

(D) Equation of the plane is

$$\begin{vmatrix} x-0 & y-7 & z+7 \\ -1 & -4 & 5 \\ -3 & 2 & 1 \end{vmatrix} = 0$$

100. (A) Find the foot of perpendicular of P(1,0,1)

$$\text{on } \frac{x-1}{1} = \frac{y-2}{0} = \frac{z-4}{-1} \text{ then find distance}$$

(B) same as (A)

$$(C) \vec{a} \cdot \vec{a} = \vec{b} \cdot \vec{b} = \vec{c} \cdot \vec{c} = 1 \text{ and}$$

$$\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{c} = \vec{c} \cdot \vec{a} = \frac{\sqrt{3}}{2}$$

$$V = \frac{1}{6} \begin{vmatrix} \vec{a} & \vec{b} & \vec{c} \end{vmatrix}$$

$$V^2 = \frac{1}{36} \begin{vmatrix} \vec{a} & \vec{b} & \vec{c} \\ \vec{a} & \vec{b} & \vec{c} \\ \vec{a} & \vec{b} & \vec{c} \end{vmatrix}^2$$

$$= \frac{1}{36} \begin{vmatrix} \vec{a} \cdot \vec{a} & \vec{a} \cdot \vec{b} & \vec{a} \cdot \vec{c} \\ \vec{b} \cdot \vec{a} & \vec{b} \cdot \vec{b} & \vec{b} \cdot \vec{c} \\ \vec{c} \cdot \vec{a} & \vec{c} \cdot \vec{b} & \vec{c} \cdot \vec{c} \end{vmatrix}$$

$$(D) (\vec{a} - \vec{c}) \cdot (\vec{b} + \vec{c}) = 0$$

$$\vec{a} \cdot \vec{b} + \vec{c} \cdot (\vec{a} - \vec{b}) = |\vec{c}|^2$$

$$(\vec{a} - \vec{b}) \cdot (\lambda \vec{a} + \mu \vec{b} + \omega (\vec{a} \times \vec{b})) = 1$$

$$\Rightarrow \lambda - \mu = 1$$

$$\Rightarrow \lambda = \mu + 1$$

$$\Rightarrow (\mu + 1)^2 + \mu^2 + \omega^2 = 1$$

$$[\vec{a} \ \vec{b} \ \vec{c}] = 0 \text{ \& \ } \vec{a} \times \vec{b} \neq \vec{0}$$

$$(\vec{c} \times \vec{b}) \cdot (\vec{a} \times \vec{b}) = \begin{vmatrix} \vec{c} \cdot \vec{a} & \vec{c} \cdot \vec{b} \\ \vec{b} \cdot \vec{a} & \vec{b} \cdot \vec{b} \end{vmatrix}$$

$$(\vec{a} \times \vec{c}) \cdot (\vec{a} \times \vec{b}) = \begin{vmatrix} \vec{a} \cdot \vec{a} & \vec{a} \cdot \vec{b} \\ \vec{c} \cdot \vec{a} & \vec{c} \cdot \vec{b} \end{vmatrix}$$

$$101. (A) |\vec{a}| = |\vec{b}| = |\vec{c}| = \sqrt{3}$$

$$\Rightarrow \text{circumference } \Delta ABC \text{ is } \vec{0}$$

$$\text{centroid} = \vec{ob} = \frac{\vec{a} + \vec{b} + \vec{c}}{3}$$

Centroid divides $\vec{OS}$ in the ratio 2:1

$$(C) \vec{x} + \vec{y} + \vec{z} = \vec{a}$$

$$\Rightarrow \vec{a} \cdot \vec{x} + \vec{a} \cdot \vec{y} + \vec{a} \cdot \vec{z} = |\vec{a}|^2 = 4$$

$$\Rightarrow \frac{3}{2} + \frac{7}{4} + \vec{a} \cdot \vec{z} = 4$$

$$\Rightarrow \vec{a} \cdot \vec{z} = \frac{3}{4}$$

$$\vec{x} \cdot (\vec{x} + \vec{y} + \vec{z}) = \vec{x} \cdot \vec{a} = \frac{3}{2}$$

$$\Rightarrow \vec{x} \cdot \vec{y} + \vec{x} \cdot \vec{z} = \frac{1}{2} \quad \text{etc.}$$

$$(D) \quad 3\vec{a} + 7\vec{b} = \lambda\vec{c} \quad \rightarrow \quad (1)$$

$$3\vec{b} + 2\vec{c} = \mu\vec{a} \quad \rightarrow \quad (2)$$

$$(1) \times 3 - (2) \times 7$$

$$\Rightarrow 9\vec{a} - 14\vec{c} = 3\lambda\vec{c} - 7\mu\vec{a}$$

$$\Rightarrow 9 = -7\mu \quad \& \quad -14 = 3\lambda$$

$$\lambda = -\frac{14}{3}, \quad -\lambda = \frac{14}{3} \quad \text{in} \quad \dots\dots\dots(1)$$

INTEGER QUESTIONS

102. Conceptual

103. Let $\vec{a}_1, \vec{a}_2, \dots, \vec{a}_{2006}$ be 2006 vectors

$$\vec{a}_1 + \vec{a}_2 + \dots + \vec{a}_{2005} = \lambda\vec{a}_{2006}$$

$$\vec{a}_2 + \vec{a}_3 + \dots + \vec{a}_{2006} = \lambda\vec{a}_1$$

$$(2005)(\vec{a}_1 + \vec{a}_2 + \dots + \vec{a}_{2006})$$

$$= \lambda(\vec{a}_1 + \vec{a}_2 + \dots + \vec{a}_{2006})$$

$$(2005)(\vec{s}) = \lambda\vec{s}$$

$$\Rightarrow (\lambda - 2005)\vec{s} = 0$$

$$\Rightarrow |\lambda - 2005||\vec{s}| = 0$$

$$|\vec{s}| = 0$$

104. Conceptual

105. We have

$$\vec{V}_1 \cdot \vec{V}_2 = a + b + c = 0$$

but $a, b, c \in \{-2, -1, 0, 1, 2\}$

now (i) if $a = 1, b = -1, c = 0$, number = $3! = 6$

(ii) if $a = 2, b = -2, c = 0$, number = $3! = 6$

(iii) if $a = 1, b = 1, c = -2$, number = $\frac{3!}{2!} = 3$

(iv) if $a = -1, b = -1, c = 2$, number = $\frac{3!}{2!} = 3$

$\therefore$ Total = 18

$$106. \quad \vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{c} = \vec{c} \cdot \vec{a} = \cos \theta$$

$$\vec{a} \cdot \vec{d} = \vec{b} \cdot \vec{d} = \vec{c} \cdot \vec{d} = \cos \alpha$$

Since $\vec{a}, \vec{b}, \vec{c}$ have to be non-coplanar.

if $\vec{d} = \alpha\vec{a} + \beta\vec{b} + \gamma\vec{c}$, by symmetry

$$\alpha = \beta = \gamma = k$$

$$\therefore \vec{d} = k(\vec{a} + \vec{b} + \vec{c})$$

$$\vec{d} \cdot \vec{a} = k(1 + \vec{b} \cdot \vec{a} + \vec{c} \cdot \vec{a})$$

$$\cos \alpha = k(1 + 2 \cos \theta) \dots\dots (1)$$

$$\vec{d} \cdot \vec{d} = 3K \cos \alpha \Rightarrow 3K \cos \alpha = 1 \dots\dots\dots (2)$$

from (1) and (2) $3 \cos^2 \alpha = 1 + 2 \cos \theta$

$$\Rightarrow 3 \left(\frac{1 + \cos 2\alpha}{2} \right) = 1 + 2 \cos \theta$$

$$\Rightarrow 3 + 3 \cos 2\alpha = 2 + 4 \cos \theta$$

$$\Rightarrow 4 \cos \theta - 3 \cos 2\alpha = 1$$

107. Let $\vec{c} = x\vec{a} + y\vec{b}$ Where x, y are scalars

$$\Rightarrow \vec{c} = x(\hat{i} - \hat{j} + 2\hat{k}) + y(2\hat{i} - \hat{j} + \hat{k})$$

$$\Rightarrow \vec{c} = \hat{i}(x + 2y) + \hat{j}(-x - y) + \hat{k}(2x + y)$$

$$\text{But } \vec{c} \cdot \vec{a} = 0$$

$$6x + 5y = 0$$

$$\Rightarrow y = \frac{-6x}{5}$$

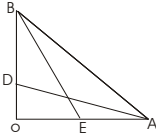
$$\text{so, } \vec{c} = \frac{-7x}{5}\hat{i} + \frac{x}{5}\hat{j} + \frac{4x}{5}\hat{k}$$

$$\text{We have } \frac{49x^2 + x^2 + 16x^2}{25} = 1 \Rightarrow x^2 = \frac{25}{66}$$

$$\therefore \vec{c} = \pm \frac{5}{\sqrt{66}} \left(\frac{-7}{5}\hat{i} + \frac{1}{5}\hat{j} + \frac{4}{5}\hat{k} \right)$$

$$p = |\vec{c} \cdot \hat{b}| = \frac{\sqrt{11}}{6} \quad \text{so } = \frac{\sqrt{11}}{p} = 6$$

108. Without loss of generality, let the right angled $\triangle OAB$ be such that $OA = OB = 1$ unit. Along OA take unit vector as $\hat{i}$ and along OB take unit vector as $\hat{j}$



$$\text{So, } \overrightarrow{OA} = \hat{i} \quad \overrightarrow{OB} = \hat{j}$$

$$\overrightarrow{AD} = \overrightarrow{OD} - \overrightarrow{OA} = \frac{j}{2} - \hat{i}$$

$$\overrightarrow{BE} = \overrightarrow{OE} - \overrightarrow{OB} = \frac{\hat{i}}{2} - \hat{j}$$

Let θ be the acute angle between the medians

$$\cos \theta = \frac{|\overrightarrow{AD} \cdot \overrightarrow{BE}|}{|\overrightarrow{AD}| |\overrightarrow{BE}|} = \frac{1}{\left(\frac{5}{4}\right)} = \frac{4}{5}$$

$$\therefore 4 \sec \theta = 5$$

109. Let $\vec{c}$ be inclined at angles θ with $\vec{a}$ and 2θ with $\vec{b}$

$$\vec{a} + \vec{b} = \vec{c} \Rightarrow \vec{a} \cdot \vec{a} + \vec{a} \cdot \vec{b} = \vec{a} \cdot \vec{c} \text{ and}$$

$$\vec{b} \cdot \vec{a} + \vec{b} \cdot \vec{b} = \vec{b} \cdot \vec{c}$$

$$\therefore 6 + 4 \cos 3\theta = c \cos \theta$$

$$6 \cos 3\theta + 4 = c \cos 2\theta$$

Eliminating c ,

$$(6 + 4 \cos 3\theta) \cos 2\theta = (6 \cos 3\theta + 4) \cos \theta$$

$$\therefore 4(4 \cos 3\theta \cos 2\theta - \cos \theta) = 6(\cos 3\theta - \cos 2\theta)$$

$$2(\cos 3\theta \cos 2\theta - \cos(3\theta - 2\theta))$$

$$= 3(\cos 3\theta \cos \theta - \cos \theta - \cos(3\theta - \theta))$$

$$\therefore 2 \sin 2\theta = 3 \sin \theta \Rightarrow \cos \theta = \frac{3}{4}$$

$$(i) \Rightarrow c = \frac{6 + 4 \cos 3\theta}{\cos \theta} = \frac{8}{3}(3 + 2 \cos 3\theta)$$

$$= \frac{8}{3}(3 + 2(4 \cos^3 \theta - 3 \cos \theta))$$

$$= \frac{8}{3} \left[3 + 2 \left(\frac{21}{16} - \frac{9}{4} \right) \right] = 5$$

110. $\vec{V}$ is coplanar with $\vec{V}_1$ and $\vec{V}_2$ and perpendicular to $\vec{V}_3$

$$\text{Let } \vec{V} = \lambda (\vec{V}_1 \times \vec{V}_2) \times \vec{V}_3$$

$$= -\lambda [\vec{V}_3 \times (\vec{V}_1 \times \vec{V}_2)]$$

$$= -\lambda [(\vec{V}_3 \cdot \vec{V}_2) \vec{V}_1 - (\vec{V}_3 \cdot \vec{V}_1) \vec{V}_2]$$

$$= -\lambda [-5(i + j - 2k) - (-2)(\hat{i} - 2\hat{j} + \hat{k})]$$

$$\vec{V} = \lambda [3\hat{i} + 9\hat{j} - 12\hat{k}] \dots\dots\dots(i)$$

projection of $\vec{V}$ on $\hat{i} - \hat{j} + \hat{k}$

$$= \frac{|\lambda[3 - 9 - 12]|}{\sqrt{3}} = 6\sqrt{3} \Rightarrow \lambda = \pm 1$$

$$\vec{V} = \pm [3\hat{i} + 9\hat{j} - 12\hat{k}] = \pm (\hat{i} + 3\hat{j} - 4\hat{k})$$

111. Projection of $A(\vec{a})$ in the plane $\vec{r} \cdot \hat{n} = 2$ is

$$\vec{a} + (2 - \vec{a} \cdot \hat{n}) \hat{n}$$

112. Conceptual

113. Let the angle between $\vec{a}$ and $\vec{b}$ is α and $\vec{a} \times \vec{b}$ and $\vec{c}$ is β

$$\therefore |[\vec{a} \vec{b} \vec{c}]| = 6 \Rightarrow \sin \alpha \cos \beta = 1$$

$$\text{so } \Rightarrow \sin \alpha = 1, \cos \beta = 1 \Rightarrow \alpha = 90^\circ, \beta = 0^\circ$$

$\Rightarrow \vec{a}, \vec{b}, \vec{c}$ are mutually perpendicular

$$\text{again } [\vec{b} \vec{c} \vec{d}] = 0 \Rightarrow \begin{vmatrix} 4 & 0 & 1 \\ 0 & 9 & \vec{c} \cdot \vec{d} \\ 1 & \vec{c} \cdot \vec{d} & 1 \end{vmatrix} = 0$$

$$\Rightarrow \vec{c} \cdot \vec{d} = \pm \frac{3\sqrt{3}}{2}$$

we have $\vec{a} \cdot \vec{d} = 0$

$$|\vec{a} \times \vec{c} \cdot \vec{d}|^2 = \begin{vmatrix} 1 & 0 & 0 \\ 0 & 9 & \frac{3\sqrt{3}}{2} \\ 0 & \frac{3\sqrt{3}}{2} & 1 \end{vmatrix}$$

$$= 9 - \frac{27}{4} = \frac{9}{4}$$

$$|(\vec{a} \times \vec{c}) \cdot \vec{d}|^2 = |(\vec{a} \cdot \vec{d}) \vec{c} - (\vec{c} \cdot \vec{d}) \vec{a}|^2$$

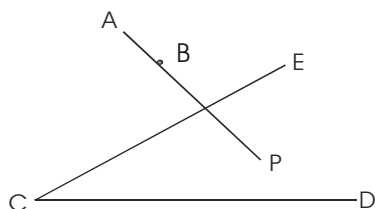
$$= \left| \frac{3\sqrt{3}}{2} \vec{a} \right|^2 = \frac{27}{4}$$

$$\text{so } |\vec{a} \times \vec{c} \cdot \vec{d}|^2 + |(\vec{a} \times \vec{c}) \times \vec{d}|^2 = \frac{36}{4} = 9$$

$$\begin{aligned} 114. (\vec{a} \times \vec{b}) \times (\vec{r} \times \vec{c}) &= [\vec{a}\vec{b}\vec{c}] \vec{r} - [\vec{a}\vec{b}\vec{r}] \vec{c} \\ &= [\vec{a}\vec{r}\vec{c}] \vec{b} - [\vec{b}\vec{r}\vec{c}] \vec{a} \\ \Rightarrow [\vec{a}\vec{b}\vec{c}] \vec{r} &= [\vec{a}\vec{b}\vec{r}] \vec{c} + [\vec{b}\vec{c}\vec{r}] \vec{a} + [\vec{c}\vec{a}\vec{r}] \vec{b} \\ \text{Now } (\vec{a} \times \vec{b}) \times (\vec{r} \times \vec{c}) &+ (\vec{b} \times \vec{c}) \\ \times (\vec{r} \times \vec{a}) &+ (\vec{c} \times \vec{a}) \times (\vec{r} \times \vec{b}) \\ &= [\vec{a}\vec{b}\vec{c}] \vec{r} - [\vec{a}\vec{b}\vec{r}] \vec{c} + [\vec{b}\vec{c}\vec{a}] \vec{r} \\ &- [\vec{b}\vec{c}\vec{r}] \vec{a} + [\vec{c}\vec{a}\vec{b}] \vec{r} - [\vec{c}\vec{a}\vec{r}] \vec{b} \\ &= 3[\vec{a}\vec{b}\vec{c}] \vec{r} - [\vec{a}\vec{b}\vec{c}] \vec{r} = 2[\vec{a}\vec{b}\vec{c}] \vec{r} \end{aligned}$$

So $\lambda = 2$

115.



$\vec{AB} = \vec{P.V.}$ of B-P.V. of A = $\hat{i} - \hat{j} + \hat{k}$
vector equation of line AB is

$\vec{r} = (\hat{i} + 2\hat{j} + \hat{k}) + t(\hat{i} - \hat{j} + \hat{k})$, where 't' is

scalar = $(1+t)\hat{i} + (2-t)\hat{j} + (1+t)\hat{k}$

$$\vec{CE} = 4\hat{i} + 5\hat{j} - 2\hat{k}$$

$$\vec{CD} = 2\hat{i} + 2\hat{j} - 2\hat{k}$$

$$\vec{CP} = (1+t)\hat{i} + (2-t+4)\hat{j} + (1+t-4)\hat{k}$$

since $\vec{CE}, \vec{CD}, \vec{CP}$ are coplanar

$$\text{so, } [\vec{CE} \vec{CD} \vec{CP}] = 0 \Rightarrow t = 2$$

so P.V. of P is $3\hat{i} + 0\hat{j} + 3\hat{k}$

$$\therefore \alpha + \beta + \gamma = 6$$

$$116. |\vec{a}| = |\vec{b}| = 1 \quad \vec{a} \cdot \vec{b} = 0$$

Let

$$\vec{l} = (\vec{a} \times \vec{b}) \times (\vec{a} - 2\vec{b}) = (\vec{a} \times \vec{b}) \times \vec{a} - 2(\vec{a} \times \vec{b}) \times \vec{b}$$

$$= |\vec{a}|^2 \vec{b} - (\vec{a} \cdot \vec{b}) \vec{a} - 2(\vec{a} \cdot \vec{b}) \vec{b} + 2|\vec{b}|^2 \vec{a}$$

$$= \vec{b} + 2\vec{a}$$

$$(2\vec{a} + \vec{b}) \cdot \vec{l} = |2\vec{a} + \vec{b}|^2 = 5$$

$$117. [\vec{a} \vec{b} \vec{c}] = 0 \Rightarrow [\vec{a} \vec{b} \vec{c}]^2 = 0$$

$$\begin{vmatrix} \vec{a} \cdot \vec{a} & \vec{a} \cdot \vec{b} & \vec{a} \cdot \vec{c} \\ \vec{b} \cdot \vec{a} & \vec{b} \cdot \vec{b} & \vec{b} \cdot \vec{c} \\ \vec{c} \cdot \vec{a} & \vec{c} \cdot \vec{b} & \vec{c} \cdot \vec{c} \end{vmatrix} = \begin{vmatrix} 1 & \cos \gamma & \cos \beta \\ \cos \gamma & 1 & \cos \alpha \\ \cos \beta & \cos \alpha & 1 \end{vmatrix} = 0$$

$$\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma - 2 \cos \alpha \cos \beta \cos \gamma = 1$$

$$118. \begin{vmatrix} -1 & c & b \\ c & -1 & a \\ b & a & -1 \end{vmatrix} = 0, a^2 + b^2 + c^2 - 2abc = 1$$

$$(b+ca)^2 = 1 - a^2 - c^2 + c^2 a^2 = (1-a^2)(1-c^2)$$

$$a^2 \leq 1 \Rightarrow c^2 \leq 1 \text{ likewise } b^2 \leq 1$$

$$119. \text{ Give } |\vec{b} - \vec{a}| = 12, |\vec{c}| = 6$$

Equation of CD is $\vec{r} = t\vec{c}$

and Equation of AB is

$$\vec{r} = \vec{a} + s(\vec{b} - \vec{a})$$

$$S.D = \frac{[\vec{a} \vec{b} - \vec{a} \vec{c}]}{|\vec{b} - \vec{a}| \times |\vec{c}|} = 8$$

$$[\vec{a} \vec{c} \vec{b}] = 8 |(\vec{b} - \vec{a}) \times \vec{c}|$$

$$6V = 8 \frac{|\vec{c}| |\vec{b} - \vec{a}|}{2}$$

$$V = \frac{8 \times 6 \times 12}{6 \times 2} = 48$$

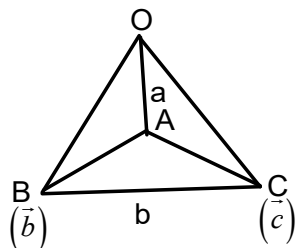
$$\frac{1}{6} V = 8$$

$$120. \text{ Let } OA = a, BC = b$$

equation of OA is $\vec{r} = t\vec{a}$

equation of BC is $\vec{r} = (1-s)\vec{b} + s\vec{c}$

$$\vec{r} = \vec{b} + s(\vec{c} - \vec{b})$$



Now S.D = d

$$\frac{[\vec{b} \ \vec{c} - \vec{b} \ \vec{c}]}{|\vec{c} - \vec{b} \times \vec{a}|} = d$$

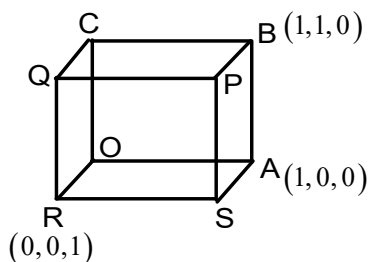
$$\frac{\vec{b} \times (\vec{c} - \vec{b}) \cdot \vec{a}}{|\vec{c} - \vec{b}| |\vec{a}| \sin \theta} = d$$

$$\frac{[\vec{a} \ \vec{b} \ \vec{c}]}{ab \sin \theta} = d$$

$$[\vec{a} \ \vec{b} \ \vec{c}] = abd \sin \theta$$

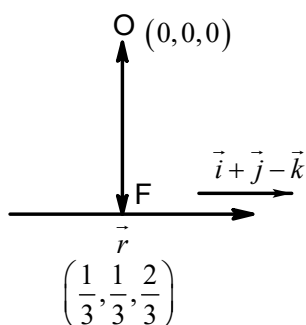
$$\text{volume} = \frac{1}{6} [\vec{a} \ \vec{b} \ \vec{c}] = \frac{1}{6} abd \sin \theta$$

121. Consider a corner O diagonal BR



equation of BR is

$$\vec{r}(1-t)\vec{k} + t(\vec{i} + \vec{j})$$



$$\overrightarrow{OF} \cdot \vec{i} + \vec{j} - \vec{k} = 0$$

$$\vec{r} \cdot \vec{i} + \vec{j} - \vec{k} = 0$$

$$t\vec{i} + t\vec{j} + (1-t)\vec{k} \cdot \vec{i} + \vec{j} - \vec{k} = 0$$

$$3t = 1 \Rightarrow t = \frac{1}{3}$$

$$OF = \sqrt{\frac{1}{9} + \frac{1}{9} + \frac{4}{9}} = \sqrt{\frac{6}{9}} = \sqrt{\frac{2}{3}} = k$$

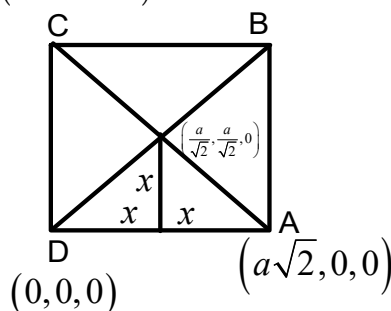
122. $x^2 + x^2 = a^2$

$$x = \frac{a}{\sqrt{2}}$$

$$2x = a\sqrt{2}$$

When it is folded along $\overline{AC}$
right angled to BAC

$$(0, a\sqrt{2}, 0)$$



B goes to $B' \rightarrow$ coordinates of $B' \rightarrow B'$

$$\left(\frac{a}{\sqrt{2}}, \frac{a}{\sqrt{2}}, 0 \right)$$

Eg: $\overrightarrow{DC} \rightarrow \vec{r} = t(a\sqrt{2}\vec{j}) \rightarrow (1) A(a\sqrt{2}, 0, 0)$

Eg.

$$\overrightarrow{AB'} \rightarrow \vec{r} = a\sqrt{2}\vec{i} + s\left(\frac{-a}{\sqrt{2}}\vec{i} + \frac{-a}{\sqrt{2}}\vec{j} + a\vec{k}\right) \rightarrow (2)$$

then S.D. between (1) & (2) is

$$\frac{\left[a\sqrt{2}\vec{i} \ \frac{-a}{\sqrt{2}}\vec{i} + \frac{a}{\sqrt{2}}\vec{j} + a\vec{k} \ a\sqrt{2}\vec{i} \right]}{\left| a\sqrt{2}\vec{i} \times \left(\frac{-a}{\sqrt{2}}\vec{i} + \frac{a}{\sqrt{2}}\vec{j} + a\vec{k} \right) \right|}$$

$$= \frac{\begin{vmatrix} a\sqrt{2} & 0 & 0 \\ \frac{-a}{\sqrt{2}} & \frac{a}{\sqrt{2}} & a \\ 0 & \frac{a}{\sqrt{2}} & 0 \end{vmatrix}}{\begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 0 & \frac{a}{\sqrt{2}} & 0 \\ \frac{-a}{\sqrt{2}} & \frac{a}{\sqrt{2}} & a \end{vmatrix}} = a\sqrt{2}(-\sqrt{2}a^2) = 2a^3$$

$$= \frac{2a^3}{\sqrt{3}.a^2} = \frac{2a}{\sqrt{3}}$$

123. $\vec{a} \times \vec{x} = \vec{x} - \vec{b}$

$$\vec{a} \times (\vec{a} \times \vec{x}) = \vec{a} \times \vec{x} - \vec{a} \times \vec{b}$$

$$(\vec{a} \cdot \vec{x})\vec{a} - (\vec{a} \cdot \vec{a})\vec{x} = \vec{x} - \vec{b} - \vec{a} \times \vec{b}$$

$$(\vec{a} \cdot \vec{x})\vec{a} = \angle \vec{x} - \vec{b} - \vec{a} \times \vec{b} \rightarrow (1)$$

$$\frac{\vec{x} \cdot \vec{a}}{|\vec{a}|} = 2 \Rightarrow \vec{x} \cdot \vec{a} = 2$$

$$2\vec{a} + \vec{b} + \vec{a} \times \vec{b} = 2\vec{x}$$

$$\Rightarrow \vec{x} = \frac{(2\vec{a} + \vec{b} + \vec{a} \times \vec{b})}{2}$$

$$\cos(90 - \alpha) = \frac{-3}{5} = \sin \alpha$$

$$\sin \alpha = \frac{-3}{5}$$

$$\sin \alpha = \frac{4}{5}$$

$$\cos \beta = \frac{2+2}{\sqrt{5} \cdot \sqrt{5}} = \frac{4}{5}$$

$$\sin \beta = \frac{3}{5}$$

$$\cos^2 \alpha + \sin^2 \beta = \frac{16}{25} + \frac{9}{25} = \frac{25}{25} = 1$$

124. Conceptual

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